Chang Liu

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Education

School of Computer Science

Carnegie Mellon University, Ph.D. Student in Machine Learning 08/2023 - 05/2028(est.)

- Advisor: Prof. Artur Dubrawski.
- Research Interests: LLM post-training, Agentic AI, machine learning in healthcare.
- Skills: LLM fine-tuning; Prompt engineering.
- Courses: Advanced Introduction to Machine Learning, Intermediate Statistics, Advanced Machine Learning: Theory and Methods, Advanced Deep Learning, Machine Learning for Large Datasets.

Yao Class, Institute for Interdisciplinary Information Sciences (IIIS) Established by Prof. Andrew C. Yao

Tsinghua University, B.Eng. in Computer Science 08/2019 - 06/2023

- GPA: **3.91/4.00**.
- TOEFL: **120/120**. Reading: 30, Listening: 30, Speaking: 30, Writing: 30.
- GRE: **340/340**. Quant: 170, Verbal: 170, Writing: 5.
- Mathematics Courses: Calculus, Linear Algebra, Abstract Algebra, Mathematics for Computer Science, Mathematics for Artificial Intelligence, Probability and Statistics.
- Computer Science Courses: Machine Learning, Reinforcement Learning, Computational Biology, Computer Vision, Deep Learning, Natural Language Processing, Introduction to Databases, Data Mining, Quantum Computer Science, Introduction to Robotics, Algorithm Design, Theory of Computation.

Industry Experience

TikTok: Video Recommendation

Research Intern 05/2026 - 08/2026

- Model user preference on PGC (professionally generated content) to boost recommendation performance.

LLM Research Experience

Experiential Abstractions in LLMs 05/2026 - Now

LLM post-training. Reinforcement learning. verl. Retrieval-augmented LLMs. Carnegie Mellon University

- Develop a novel framework for improving LLM problem-solving abilities by first extracting abstractions from training, and then absorbing them via in-context learning or RL. (Build on ByteDance's verl library for RL.)
- Improve mathematical reasoning abilities compared to naive GRPO, even without a strong teacher for abstraction extraction.
- Accepted at EMNLP Findings. Paper link here.

Lightweight LLM agents for MLE 05/2025 - 08/2025

Coding agents. AWS strands. Prompt engineering. Scaffold engineering Carnegie Mellon University

- Developed a lightweight LLM agent for machine learning engineering (MLE).
- Adopt the CodeAct paradigm (draft and revise) and invoke mini-swe-agent as a tool.
- Use AWS's strands library for building the agents.
- Project link here.

Depth pruning and emergent abilities of pruned LLMs 12/2024 - 09/2025

LLM pruning. LLM inference. Huggingface. Benchmarking Carnegie Mellon University

- Proposed a novel depth-based LLM pruning methodology.
- Systematically benchmarked the emergent abilities of aggressively pruned LLMs.
- Paper link here.

Healthcare Research Experience

Multimodal outbreak detection in hospitals 10/2023 - Now

Multimodal ML. ML for healthcare. Carnegie Mellon University

- Develop a multimodal machine learning framework for hospital outbreak detection that integrates signals from MALDI-TOF mass spectra, antimicrobial resistant patterns, and electronic health records (EHR).
- Enable cost-effective outbreak detection that can reduce dependency on whole genome sequencing gold standard.

Multimodal disease target identification through a probabilistic knowledge graph 09/2021 - 05/2023

Multimodal ML. Graph Neural Networks. Computational biology. Tsinghua University

- Develop a novel method of augmenting biological networks with literature evidence to construct a probabilistic knowledge graph.
- Develop a graph neural network to predict target candidates from the knowledge graph, achieving superior performance to state-of-the-art models in terms of accuracy (esp. on sparse data) and literature support for top novel predictions.
- Conduct bioinformatics analyses and cooperate with experimental validation of the identified colorectal cancer and melanoma targets.

Publications

1. Alexander J. Sundermann*; Jieshi Chen; **Chang Liu**;...; Artur Dubrawski; Lora Lee Pless; Lee H. Harrison. "MALDI-TOF for healthcare transmission prediction: comparison with whole genome sequencing" *Infection Control & Hospital Epidemiology*, October 2026.

2. **Chang Liu**; Xinyu Li; Artur Dubrawski*."Notes to Self: Can LLMs Benefit from Experiential Abstractions?" *EMNLP Findings*, August 2026. paper link here.

3. **Chang Liu**; Jieshi Chen; Alexander J. Sundermann; Kathleen Shutt; Marissa P. Griffith; Lora Lee Pless; Lee H. Harrison; Artur Dubrawski*."Towards Practical Multimodal Outbreak Detection," *AMIA Annual Symposium*, May 2026.

4. **Chang Liu**; Jieshi Chen; Alexander J. Sundermann; Kathleen Shutt; Marissa P. Griffith; Lora Lee Pless; Lee H. Harrison; Artur Dubrawski*."Exploring the Utility of MALDI-TOF Mass Spectrometry and Antimicrobial Resistance in Hospital Outbreak Detection," *AMIA Amplify Informatics Conference*, February 2026.

5. **Chang Liu**†; Arjun Choudhry†; Yifu Cai; Nina Żukowska; Mononito Goswami; Artur Dubrawski. "Depth as a Scaling Vector: Simple Pruning and Evaluation of Emergent Abilities in Pruned LLMs," *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*, September 2025.

6. Arjun Choudhry†; **Chang Liu**†; Nina Żukowska; Yifu Cai; Mononito Goswami; Artur Dubrawski. "LayerMerge: Modality-Agnostic Depth Pruning for Efficient Foundation Model Deployment," *NeurIPS 2025 Workshop on Efficient Reasoning*, September 2025.

7. Mikołaj Piórczyński; Wojciech Łapacz; Xinyu Li; **Chang Liu**; Abby Turner; Artur Dubrawski. "From Aggregation to Guidance: Strategies for Personalized Federated Fine-Tuning of Foundation Models" *NeurIPS 2025 Workshop on Unifying Representations in Neural Models*, September 2025.

8. **Chang Liu**; Jieshi Chen; Lee H. Harrison; Artur Dubrawski*. "Bridging the Utility Gap Between MALDI-TOF and WGS for Affordable Outbreak Cluster Detection," *The AHLI Conference on Health, Inference, and Learning (CHIL)*, April 2025.

9. **Chang Liu**; Jieshi Chen; Lee H. Harrison; Artur Dubrawski*. "Multimodal Structure Preservation Learning,"*arXiv preprint*, October 2024.

10. **Chang Liu**†; Kaimin Xiao†; Cuinan Yu†; Yipin Lei†;...; Dan Zhao*; Fengfeng Zhou*; Haidong Tang*; Jianyang Zeng*. "A Probabilistic Knowledge Graph Approach for Target Identification," *PLOS Computational Biology*, April 2024.

11. **Chang Liu**†; Cuinan Yu†; Yipin Lei†;...; Dan Zhao*; Fengfeng Zhou*; Jianyang Zeng*. "Improving Target-disease Association Prediction through a Graph Neural Network with Credibility Information," proceedings of the *Pacific Symposium on Biocomputing*, January 2023.

Honors & Awards

- **Comprehensive Merit Award (7/32)**, *Tsinghua University* 2022
- **Comprehensive Merit Award (6/32)**, *Tsinghua University* 2021
- **Excellence Award for Volunteering Services**, *Tsinghua University* 2020
- **Freshmen Scholarship**, *Tsinghua University* 2019
- **University Full Scholarship for Future Scholars**, *Tsinghua University* 2019

Academic Service

- **Peer Review:** Reviewed 19 academic papers. NeurIPS Neural Artifacts Workshop 2026 (3), NeurIPS 2026 (1), NeurIPS UniReps Workshop 2025 (2), AMIA Amplify Informatics Conference 2026 (5), AMIA Annual Symposium (8).